

Trial Test 5

- Submit your answers via Google Forms (you have only 1 hour and 10 mins to complete).
- Your score will be available immediately after the test. Please, check your emails.
- If you have technical issues, let me know in DM or in the Group Chat.
- You are not allowed to use the calculator. Also, do not google, be honest!
- Do not panic and manage your time wisely. Read the questions carefully and try to answer all of them.

Good Luck!

1) Simplify:

$$(\sqrt{8} + 2)(\sqrt{2} + 4)$$

a) $4 + 10\sqrt{2}$

b) $16 + 5\sqrt{2}$

c) $24 + 5\sqrt{2}$

d) $12 + 10\sqrt{2}$

2) Find a sum of the solutions:

$$2\left(\frac{x}{2} + 3\right)^2 - \left(\frac{x}{2} + 3\right) - 36 = 0$$

a) -14

b) -11

c) 3

d) 0

3) What is the value of x?

$$5^{x+1} * 25^{3x} = 125^{19}$$

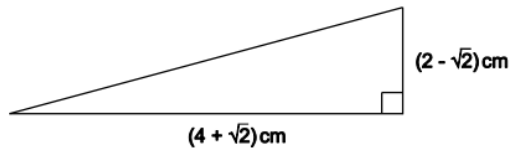
a) 3

b) 2

c) 8

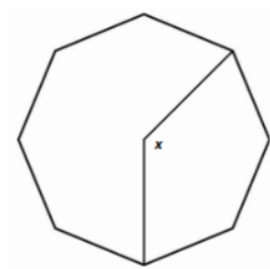
d) 9

4) The right-angled triangle shown has horizontal and vertical sides measuring $4 + \sqrt{2}$ cm and $2 - \sqrt{2}$ cm respectively.



- a) $3 - \sqrt{2}$
- b) $3 + 3\sqrt{2}$
- c) $5 - \sqrt{2}$
- d) $5 + 3\sqrt{2}$

5) The figure shown is a regular polygon.
What is the measure of angle x ? (in degrees)



- a) 120
- b) 135
- c) 145
- d) 150

6) A 120 liter solution is 25% alcohol.

How much alcohol must be added to produce a 50% alcohol solution?

a) 360

b) 240

c) 720

d) 480

7) In an arithmetic progression, the sum of first term, third term and the fifth term is 39

and the sum of second term, fourth term and the sixth term is 51.

Find the tenth term of the sequence.

a) 46

b) 39

c) 41

d) 34

8) When the expression $(2x + 3)^2 - (x - 3)^2$ written in the form $p(x + q)^2 + r$,

where p , q and r are constants, what is the value of r ?

a) 0

b) 3

c) -27

d) -9

9) A solid sphere of radius r fits inside a hollow cylinder. The cylinder has the same internal diameter and length as the diameter of the sphere. The volume of a sphere is $\frac{4}{3}\pi r^3$, where r is the radius of the sphere.

What fraction of the space inside the cylinder is taken up by the sphere?

a) $\frac{1}{4}$

b) $\frac{1}{3}$

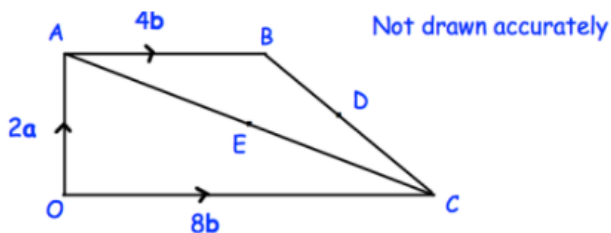
c) $\frac{1}{2}$

d) $\frac{2}{3}$

10) OABC is a trapezium. Point D is the midpoint of BC. Point E is the midpoint of AC.

Write the vector $\overrightarrow{AE}$ in terms of a and b .

$\overrightarrow{OA} = 2a$ $\overrightarrow{AB} = 4b$ and $\overrightarrow{OC} = 8b$



a) $-a+4b$

b) $-a-4b$

c) $-2a-4b$

d) $2a-4b$

11) Find the area of the sector whose arc length and radius are 16 cm and 9 cm respectively.

a) 36

b) 72

c) 64

d) 86

12) Convert 25 m/s to km/h

a) 115

b) 65

c) 75

d) 90

13) Find a sum and write it as a fraction:

$0.6(8)+0.(64)$

a) $\frac{661}{495}$

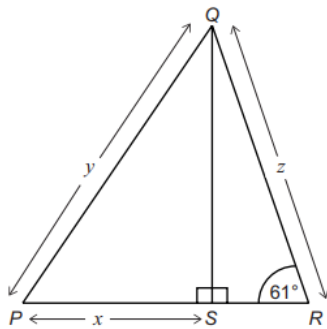
b) $\frac{741}{495}$

c) $\frac{721}{551}$

d) $\frac{641}{551}$

14) In the diagram, QS is perpendicular to PR. PS = x cm PQ = y cm QR = z cm angle QRS = 61 ° PSR is a straight line.

Which one of the following is an expression for the length z, in cm?



[diagram not to scale]

a) $\sqrt{y^2 - x^2} \sin 61$

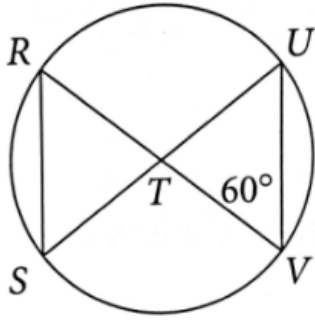
b) $\sqrt{y^2 - x^2} \cos 61$

c) $\frac{\sqrt{y^2 - x^2}}{\cos 61}$

d) $\frac{\sqrt{y^2 - x^2}}{\sin 61}$

15) The circle shown has center T. The measure of angle TVU is 60°. If the circle has a radius 3.

What is the length of segment RS?



- a) 3
- b) $2\sqrt{2}$
- c) $3\sqrt{3}$
- d) $6\sqrt{2}$

16) The number of visitors to some tourist attractions is shown in the table below.

How many more people visited the Theme Park than the Science? Write your answer in standard form.

The King's Palace	5.4 million
Castle	923,840
Theme Park	1.43×10^7
Science Museum	4,192,900

- a) 1.01071×10^8
- b) 1.01071×10^5
- c) 1.01071×10^7
- d) 1.01071×10^6

17) What is the surface area of a regular square pyramid with a slant height of 12 cm and a lateral edge of 13 cm?

- a) 240
- b) 340
- c) 600
- d) 100

18) PR and QS are the diagonals of a rhombus PQRS. $PR = (3x + 2)$ cm $QS = (8 - 2x)$ cm The area of PQRS is 11cm squared. What is the difference, in cm, between the two possible lengths of PR?

- a) 8
- b) $2\frac{2}{3}$
- c) 14
- d) $4\frac{1}{2}$

19) Two similar cones have radii of 4 cm and 6 cm. The total area of a smaller cone is 72π cm squared. Find the area of the bigger one.

- a) 162π
- b) 288π
- c) 144π
- d) 164π

20) Find the complete set of values of x for which

$$\frac{x^3 - 6x^2 + 9x - 4}{x} > 0$$

a) $0 < x < 4$

b) $0 < x < 1, x > 4$

c) $x < 0, 1 < x < 4$

d) $x < 0, x > 4$

21) If function $f(x)$ is defined as $f(x) = 1 + x^2$, then $f(\frac{x-1}{x+1}) =$

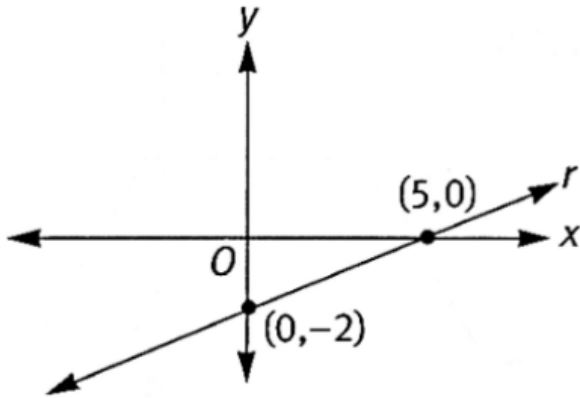
a) $\frac{x^2}{(x+1)^2}$

b) $\frac{x^2+1}{(x+1)^2}$

c) $2\frac{x^2-1}{(x+1)^2}$

d) $2\frac{x^2+1}{(x+1)^2}$

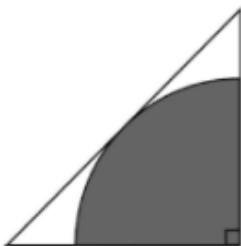
22) Line r is a straight line as shown. Which of the following points lies on line r ?



- a) (7,3)
- b) (10,2)
- c) (8,2)
- d) (9,3)

23) The diagram shows a quarter of a circle surrounded by an isosceles triangle. The radius of the circle is r .

Which one of the following expressions represents the unshaded area?

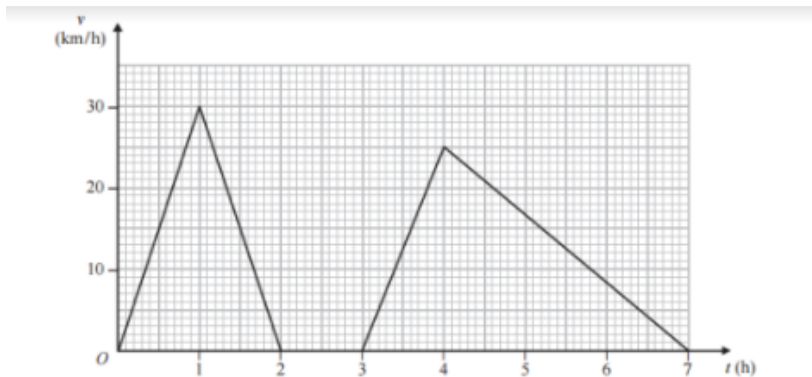


- a) $(4 - \pi)r^2$
- b) $(1 - \frac{\pi}{4})r^2$

c) $(2 - \frac{\pi}{4})r^2$

d) $r^2 - \frac{\pi r^2}{2}$

24) The speed-time graph shows information about the motion of a particle in a 7 hour period. Calculate the total distance, in km, travelled by the particle in the 7 hours.



- a) 80 km
- b) 160 km
- c) 40 km
- d) 320 km

25) The curve $y = \sin x$ is stretched by a scale factor of $\frac{1}{2}$ parallel to the x-axis and then translated by $\frac{\pi}{4}$ in the negative x-direction.

What is the equation of the new curve?

a) $y = \sin(2x + \frac{\pi}{4})$

b) $y = \sin(2x - \frac{\pi}{2})$

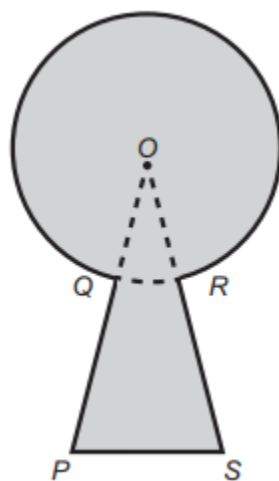
c) $y = \sin(2x - \frac{\pi}{4})$

d) $y = \sin(2x + \frac{\pi}{2})$

26) The diagram shows the outline of a keyhole consisting of three straight sides and an arc from a circle.

The sides PQ and RS are both 18 mm in length and when extended meet at the centre of the circle O forming an angle of $\frac{\pi}{6}$ radians. The longer arc from Q to R has length 22π mm.

What is the area, in mm^2 , of the keyhole as shaded in the diagram?



[diagram not to scale]

a) $132\pi + 225$

b) $132\pi + 225\sqrt{3}$

c) $144\pi + 225$

d) $144\pi + 225\sqrt{3}$

27) Angelo bought apples and bananas at the fruit stand. He bought 20 pieces of fruit and spent \$11.50. Apples cost \$0.50 and bananas cost \$0.75 each. How many apples did he buy?

a) 9

b) 7

c) 6

d) 14

28) Rearrange to give x in terms of y :

$$y = x^2 + 6x - 5$$

a) $x = \sqrt{y - 14} + 3$

b) $x = \sqrt{y - 14} - 3$

c) $x = \sqrt{y + 14} + 3$

d) $x = \sqrt{y + 14} - 3$

29) If $\diamond x$ is defined as the largest even number that is less than the negative square root of x .

What is the value of $\diamond 81$?

a) -82

b) -80

c) -10

d) -8

30) Find the LCM of 96 and 404

a) 14400

b) 2424

c) 4848

d) 9696